



SKILLS

Draw.io (Design)	5+ yrs
Microsoft 365	5+ yrs
Technical Writing	5+ yrs
LaTeX (Overleaf)	5+ yrs
Kali Linux	learning
Wireshark Analysis	learning
Python	learning
ML/DL/AI/Blockchain	learning

CONTACT

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ZAHEED AHMED BHUIYAN

RESEARCH INTERESTS

Cybersecurity, Secure Embedded Systems, AI, ML, Networking, SDN, IoT, Cloud Computing, Cloud Security, Green Computing, Healthcare Technologies, Quantum Computing, Grid Level Energy Storage Systems.

PROFILE

My academic journey has been marked by a profound fascination with cybersecurity and its vital role in our digital landscape.

While my expertise encompasses a wide array of computer science domains, I have a keen interest in other areas. Currently, I am pursuing my PhD in Information Technology at Towson University in Baltimore, Maryland, USA, and also serving as a Research Assistant and Teaching Assistant under the supervision of my PhD advisor in the Department of CIS.

PUBLICATIONS

A) Direct Contributions:

1. "An Integrated Scalable Framework for Cloud and IoT based Green Healthcare System"

Motaharul Islam and **Z. A. Bhuiyan**, "An Integrated Scalable Framework for Cloud and IoT Based Green Healthcare System," in IEEE Access, vol. 11, pp. 22266-22282, 2023. DOI: 10.1109/ACCESS.2023.3250849 ISI Indexed, Impact Factor: 3.367, **Q1**

2. "On the (in)Security of the Control Plane of SDN Architecture: A Survey"

Z. A. Bhuiyan, S. Islam, M. M. Islam, A. B. M. A. Ullah, F. Naz and M. S. Rahman, "On the (in)Security of the Control Plane of SDN Architecture: A Survey," in IEEE Access, vol. 11, pp. 91550-91582, 2023, DOI: 10.1109/ACCESS.2023.3307467. ISI Indexed, Impact Factor: 3.476, **Q1**

3. "An Empirical Research Towards the State of Transition from IPv4 to IPv6"

Z. A. Bhuiyan, A. B. M. Ahasan Ullah, M. M. Hasan Khan, S. Morshed and M. M. Islam, "An Empirical Research Towards the State of Transition from IPv4 to IPv6," 2023 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka, Bangladesh, 2023, pp. 1-6, doi: 10.1109/STI59863.2023.1

4. "Enabling Pandemic-Resilient Healthcare: NB-IoT and Edge Intelligence for Real-time Monitoring" in CAAI Transactions on Intelligence Technology [Scopus/Clarivate Indexed, Impact Factor: 5.91, **Q1**

M. M. Islam, M. K. Hasan, S. Islam, M. Balfaqih, Ahmed, N. Alalwan, N. Safie, **Z. A. Bhuiyan**, R. Thakkar and T. Ghazal: Enabling pandemic-resilient healthcare: narrowband Internet of Things and edge intelligence for real-time monitoring. CAAI Trans. Intell. Technol. 1-18 (2024). <https://doi.org/10.1049/cit2.12314>

5. "A Cyber Analysis of Threats and Countermeasures: Focusing on the GPS-NTP Synchronization System in Power Plants" (In progress)

S. E. Reza, G. J. Haque, **Z. A. Bhuiyan**, S. I. Ratan, M. M. Islam, P. H. Pham and M. M. Hassan.

Additional Tools/Software

- C (Basics)
- Java-OOP (Basics)
- Packet Tracer (Basics)
- GNS3 (Basics)
- Python (Learning)
- FTK Imager (Basics)
- Bulk-Extractor (Basics)
- Dumpit (Basics)



B) Assist:

1. "An analytical approach to error detection and correction for onboard nanosatellites"

Islam, M.M., Hasan, M. and **Bhuiyan, Z.A.**, 2023. An analytical approach to error detection and correction for onboard nanosatellites. EURASIP Journal on Wireless Communications and Networking, 2023(1), pp.1-23. DOI: 10.1186/s13638-023-02259-y [ISI Indexed, Impact Factor: 2.559, **Q2**]

C) Supervision and Guidance:

1. "Co-Pilot for Project Managers: Developing a PDF-Driven AI Chatbot for Facilitating Project Management"

K. Alam, M. Haq Bhuiyan, M. Shafiqul Islam, A. Hossain Chowdhury, **Z. Ahmed Bhuiyan** and S. Ahmmmed, "Co-Pilot for Project Managers: Developing a PDF-Driven AI Chatbot for Facilitating Project Management," in IEEE Access, vol. 13, pp. 43079-43096, 2025, doi: 10.1109/ACCESS.2025.3548519 [ISI Indexed, Impact Factor: 3.4, **Q1**]

2. "A Novel Framework for Seasonal Affective Disorder Detection: Comprehensive Machine Learning Analysis Using Multimodal Social" submitted to *Acta Psychologica* (Accepted) [Scopus, SSCI, DOAJ, PubMed Indexed, Impact Factor: 2.1, **Q1**]

M. Shamsuzzoha, T. T. B. Audry, M. J. Alam, **Z. A. Bhuiyan**, M. M. Islam and M. M. Hassan.

3. "Enhancing Android Malware Detection with Deep Learning: Comparative Analysis of ANN and Autoencoders on SMOTE-Balanced Datasets" (In progress)

T. T. B. Audry, M. J. Alam, **Z. A. Bhuiyan**, M. M. Islam and M. M. Hassan.

D) Review Works:

1. Reviewer, The Journal of Supercomputing, Springer Nature, (2024 -)

2. Reviewer, Mobile Networks and Applications, Springer Nature, (2025 -)

3. Reviewer, Discover Health Systems, Springer Nature, (2025 -)

4. Reviewer, Scientific Reports, Springer Nature, (2025 -)

WORK EXPERIENCE

Graduate TA CIS Department, Towson University	2025 -
Assisting professor conducting labs, assisting undergrad students, grading, etc.	
Graduate RA CIS Department, Towson University	2025 -
Research in the field of computer science under my PhD supervisor at Towson University.	
Graduate TA MSCSE Program, United International University	2023 - 2025
MSCSE Program Development, Designing publicity posters, Organising invited talks, New Masters Program Development, Tasks related to University OBE curriculum and accreditation, Helping graduate student groups with technical writing and research publications.	

EDUCATION

2020 - 2023

M. Sc. CSE (Cybersecurity)

United International University

My priority behind my master's studies was to get a start on understanding the foundations of cyber security, and I carefully chose the credit courses to support my goal.

Apart from choosing courses directly related to cyber security, my secondary priority was to choose other courses related to internet architectures, future internet technologies, and topics that will assist me in a deeper understanding of the security domain.

My decision-making process may give an insight into how I proceed to solve an issue.

2010 - 2015

B. Sc. EEE

Ahsanullah University of Science and Technology (AUST)

Major of my graduate studies was related to power systems. Apart from that, I had introductions to digital electronics, signal analysis, optical fiber communications, etc.



Research Assistant

Under Prof. Dr. Motaharul Islam, United International University

2022 - 2025

Research Project: An Integrated Scalable Framework for Cloud and IoT-based Green Healthcare System.

Additional Tasks: Helping undergrad student groups with research work, Occasional exam duties, etc.

Independent Consultant

FTTX Technology

2018 - 2020

Implementation/procurement of FTTX technology, Designing networks.

Executive, Product and Account Management

Network Hardware Solutions Ltd.

2015 - 2017

Handling foreign OEMs, Tender participation, Technical compliance sheet, Engaging clients during presales, FTTX solutions for passive networking etc.

Officer, Support

Network Hardware Solutions Ltd.

2015 - 2017

IT support, Troubleshooting, Testing Faulty Networking Products - RMA (Wi-Fi Routers, Network Switches).

Technologies and Brands: FTTX technology, LAN Solution, and I have worked with some brands: CISCO, Allied Telesis, EDIMAX, linkbasic, RAD, MikroTik, V Solution, etc.

EDUCATION

MSCSE, UNITED INTERNATIONAL UNIVERSITY

CGPA: 3.97 (out of 4.00), Summa cum laude

2020 - 2023

Completed Courses: Research Methodology; Cloud Computing; Digital Forensics; Cyber Crime & Cyber Terrorism; Future Internet; Internet Architecture and Protocols; Network Security; Cyber Security Standards, Governance and Management; Data Structures and Algorithms; Ethical Hacking; Object Oriented Programming; IT Audit and Assessment; Security Evaluation and Assessment Methodology; Cloud Security.

Technical Papers & Projects:

1. "Detection of Deep Fake Using Deep Learning Approach: A Review" (Term Paper)
2. "A Study of Forensic Analysis on Firewalls: A Survey" (Term Paper)
3. A Term Project utilizing different forensic tools.
4. "Trend Analysis of Credit/Debit Card attack on Bangladesh perspective" (Term Paper)
5. A Technical Presentation on "AKARI: A Newgen Network Architecture."
6. A Technical Presentation on "Wireless Mesh Networks: Focusing IEEE 802.11s Standard."
7. Designing an IT Audit and Control Assessment Using a Scenario etc.

CERTIFICATIONS

CCNA

Vendor Certification (up to 2018)

Introduction to Cybersecurity

Certificate issued by the CISCO Networking Academy under the expert guidance of Prof. Mohammad Mamun Elahi, CSE Department, United International University.

Cybersecurity Essentials

Certificate issued by the CISCO Networking Academy under the expert guidance of Prof. Mohammad Mamun Elahi, CSE Department, United International University.

NDG Linux Essentials

Certificate issued by the CISCO Networking Academy powered by NDG under the expert guidance of Prof. Mohammad Mamun Elahi, CSE Department, United International University.



B. Sc. in EEE, Ahsanullah University of Science and Technology

CGPA: 3.105 (out of 4.00)

2010 - 2015

Important Courses: Power System, Power System Protection, Biomedical Instrumentation, Telecommunication Engineering, Mobile Cellular Communication, Optical Fiber Communication, Power Electronics, Control Systems, Solid State Devices, Processing & Fabrication Technology, Microprocessor, Digital Signal Processing, Communication Theory, Digital Electronics, Signals & Linear Systems, Electrical Properties of Materials, Energy Conversion, Electronics, Programming Language (C, MATLAB), Electrical Circuits, Electrical Service Design, Mathematics etc.

Thesis & Project: A Thesis on "PLC Based Elevator Control System."

During the thesis, our group worked on an elevator system. Our goal was to add some advantages to the existing system. We used very common materials to make this system. The protection part of our system was done completely. However, the controlling part of the system was yet to be completed, and we left the later part of this project to our successive group.

CERTIFICATIONS

CCNA Routing and Switching

AIUB Institute of Continuing Education

2015

Learning Outcomes: Network fundamentals, common routing protocols, network topology design etc.

Introduction to Cybersecurity

CISCO Networking Academy

2021

Learning Outcomes: Preliminary understanding of different cyber security topics, Global implications of cyber threats, Ways in which networks are vulnerable to attack, Impact of cyber-attacks on industries, cyber security as a profession etc.

Cybersecurity Essentials

CISCO Networking Academy

2021

Learning Outcomes: Learning about different tactics, techniques and procedures used by cybercriminals, CIA principles related to cyber security countermeasures etc.

NDG Linux Essentials

CISCO Networking Academy

2021

Learning Outcomes: An introduction to Linux as an operating system, basic open source concepts and the basics of the Linux command line etc.

Technical Support Fundamentals

Google

2021

Learning Outcomes: An online non-credit course authorized by Google and offered through Coursera. In this course, we were introduced to the world of IT. I learned about the different facets of Information Technology, like computer hardware, the Internet, computer software, troubleshooting, and customer service.

Extra Curricular Activities:

Divisional Champion in the junior category in the Bangladesh Math Olympiad in 2008.

I was part of the champion team in the interdepartmental debate championship at AUST.

Was a part of -

- 1) Bangladesh National Cadet Core at the Higher Secondary level
- 2) Boy Scouts at the Secondary level
- 3) Cub Scouts at the Primary level

Language Proficiency:

Bengali (Native)

English (Professional)

Hindi (Listening & Speaking)



REFERENCES

Dr. Wassila Lalouani (PhD supervisor)

Assistant Professor, Dept. of CIS, Towson University

Contact: wlalouani@towson.edu

Dr. Motaharul Islam (Masters supervisor)

Professor, Dept. of CSE, UIU

Contact: motaharul@cse.uiu.ac.bd

Dr. Suman Ahmmed

Associate Professor & Head, Dept. of CSE, UIU

Contact: suman@cse.uiu.ac.bd